

BSP Release Note

for Windows[®] CE.NET 4.2

MMSP[™] 2

Release Note

MDMP2520FBSPRN-WC42

Ver. 2.4.2 beta / July 27, 2005

MP2520F

*Platform Builder for Microsoft Windows[®] CE.NET
for MMSP 2 Application Processor*

MAGIC EYES

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1. Release Baseline

1.1. Software Stack

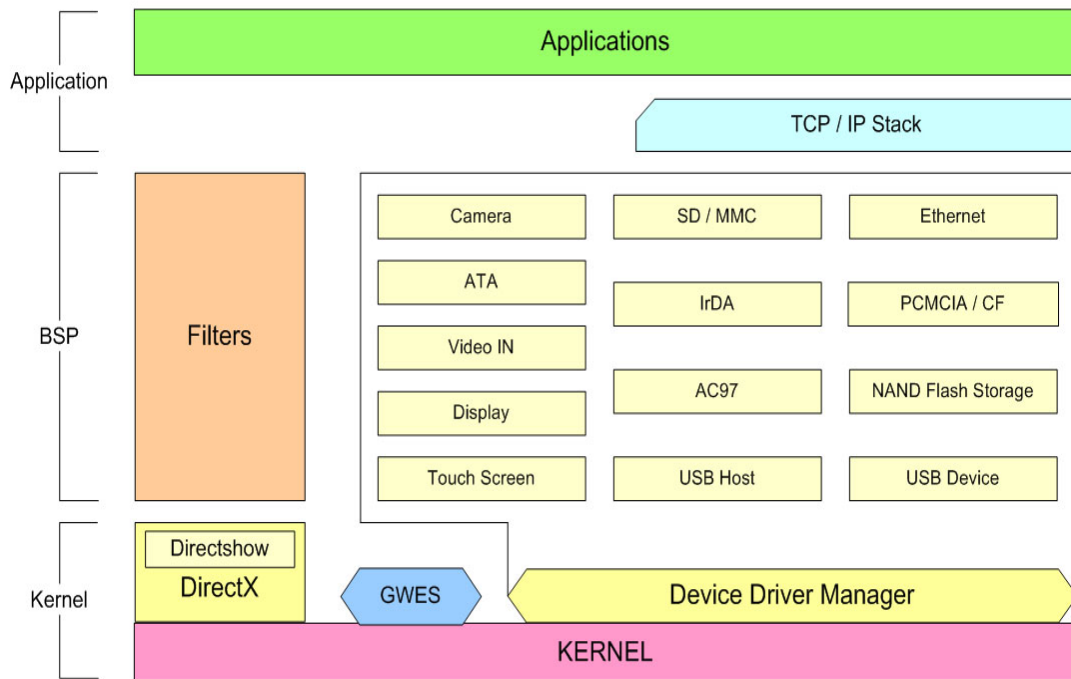


Figure 1. Software Stack

1.2. Software Modules

Basic

Kernel

The kernel is the core of the Windows CE operating system (OS). The kernel provides the base OS functionality for any Windows CE-based device. This functionality includes process, thread, and memory management. The kernel also provides some file management functionality. The kernel services enable applications to use this core functionality. MMSP2 BSP includes some kernel functions to run OS on MMSP2 based system.

Features	● OS initialization code ● interrupt handling of principal peripheral ● Power suspend/ wake-up ● Quick rebooting (boot from valid OS image in RAM without loading process) ● Remote Kernel Debugging with KITL interface
Not Supported Features	

Ethernet Bootloader

Ethernet bootloader receives Windows CE OS image (nk.bin) from platform builder through Ethernet link and load it to RAM of the MMSP2 based system. It can also store received image to the NAND flash of the system. If valid OS image is stored in NAND flash, Ethernet bootloader can load OS image from NAND flash to RAM.

Features	● Load image from Windows CE platform builder through Ethernet link ● Load image from the image which is stored in NAND flash. ● Quick booting mode support (To use this mode, OS image should be built with quick boot mode option.)
Not Supported Features	

Device Drivers

Display Driver

DirectDraw Display Driver

Features	● 8 / 16 / 24 bpp ● Output mode ■ LCD 640x480x60, 800x600x60, 800x480x60 ■ Conexant CX25874 ARGB : 640x480x60, 800x600x60 Composite/S-Video : 720x480x30 Component : 720x480x60 ■ Silicon Image SIL164 DVI 800x600x60 ● Hardware Overlay ■ 2 independent hardware overlay ■ YUV 4:2:2 format support ■ Fine or coarse scale up/down ■ Color control : brightness, contrast, hue, saturation ● Gamma correction ● Hardware cursor ● Bit-blit hardware acceleration ● OSD layer ● Sub-picture Layer ● Dynamic Rotation & Output Mode Change (GUI resolution is fixed.)
Sample Application	● APPS\COLORCTL ■ Sample application for color controls of overlay ● APPS\OSDDemo ■ Sample application for OSD layer ● APPS\SPUDemo ■ Sample application for SPU layer ● APPS\SetMode ■ Sample application for Dynamic Rotation & Output Mode Change. ● APPS\TestRGB ■ Sample application for controls of RGB Layer.

Omnivision OV9640 CMOS Sensor Driver

A device driver to support 1.3 mega pixel CMOS camera.

Feature	● Supports the OV9640 camera chip ● Get image data from camera and transfer to ISP ● Supports BayerRGB/CCIR656/CCIR656Like ● Supports VGA/QVGA/CIF/QCIF resolutions. ● Supports H/V mirrors. ● Supports to adjust a frame rate.
Sample Application	● FILTER\ISPSrc\MESCS ● APPS\CamView

Philips SAA7113 VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the SAA7713 video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs. ● Supports the TSCH6/PTI5NF Tuner
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Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn
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Ti TVP5150A VideoDecoder Driver

A device driver to support a video decoder IC

Feature	● Supports the TVP5150A video decoder chip ● Get image data from external device and transfer it to ISP ● Supports S-Video/Composite inputs.
Sample Application	● FILTER\ISPSource\MESVIS ● APPS\VideoIn

MMSP2ISP Driver

A device driver to support ISP & SCALER module of MMSP2

Feature	● Save image data from external device to memory ● Supports interfaces to set timing parameters ● Supports YUV 4:2:2 and 2D YUV 4:2:0 format
Future release	● Provides information for AF/AE/AWB control ● Provides interfaces for color controls
Sample Application	● FILTER\ISPSource\MESCS ● FILTER\ISPSource\MESVIS ● APPS\CamView ● APPS\VideoIn

I2C Driver

A device driver to support I2C interface

Feature	● Use GPIO module to make I2C signals instead of I2C module in the MMSP2
Future release	● Supports the Hardware I2C in the MMSP2

AC97 Driver

AC97 Audio Driver to play wave audio files.

Feature	● Supports REALTEK six channel AC97 audio codec ALC650 ● Compatible with AC97 2.2
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ATA Driver

A storage device driver that manages read/write operation to ATA hard disk.

Feature	● ATA Hard Disk Driver ● Support PIO / UDMA mode Default mode is PIO. UDMA mode conflicts with with MPEG H/W dec UDMA mode can be used only for file transfer
---------	--

Not Supported Features	• ATAPI is not supported.
Limitations	• Maximum partition size is limited to 136GB. • This ATA disk driver can recognize partitions in ATA HDD that is created by desktop Windows, but mounting partition created with this driver on desktop Windows can be failed. • If a HDD has multiple partitions, mounting some partition can be failed. In this case, only primary partition will be mounted.

SD/MMC Storage Driver

A storage device driver that manages read/write operation to SD/MMC memory card.

Features	• SD memory card can be mounted as a storage device whenever the system runs (hot-swap of the card is supported)
Not Supported Features	• MMC card is not supported.

NAND Flash Storage Driver

A storage device driver that maps part of NAND flash memory as a storage device.

Feature	• Can use NAND flash memory space as a storage device except of domain that stores OS image.
Limitations	• support only 8-bit bus and size of <= 128MB • The size of NAND flash memory used for storage is fixed. • One bank with 32MB(8-bit bus, 3-addr type) NAND is tested.

USB Host Driver

A driver that manages USB devices connected to MMSP2 system. The universal serial bus (USB) is an external bus architecture for connecting a USB-compatible peripheral device such as a mouse or storage device to a host computer. USB is a communication protocol that supports serial data transfers between a host computer and a USB-compatible peripheral device.

Feature	• MMSP2 System works as an USB Host when USB Devices are connected. • Recognize mouse, keyboard, and storage device through class drivers supported by Windows CE.
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USB 1.1 Function Driver

A driver that makes MMSP2 system work as a USB device.	
Feature	• USB 1.1 device controller on MMSP2 is used. • MMSP2 System works as USB device when a USB host is connected • ActiveSync link is possible with USB connection

USB 2.0 Function Driver (Mass Storage Interface)	
A driver that makes MMSP2 system work as an external mass storage when connected to a USB host.	
Feature	• NET2272 USB 2.0 device controller is used. • MMSP2 System works as USB external storage device when USB host is connected

CS8900 Ethernet Driver	
A driver that facilitates communication between CS8900 Ethernet adapter card and network protocol stack provided with Windows CE operating system.	
Feature	• Connect Local Area Network through Ethernet connection • Network features supported by Windows CE can be used. (Web browser, FTP server, etc.)

PCMCIA / CF Driver	
A driver that facilitates access service to PCMCIA/CF memory.	
Feature	• Support PCMCIA / CF storage device. (CF/PCMCIA ATA storage driver that runs on this driver is required – Provided by Microsoft)

Remote Controller Driver	
A driver that receives input from IR remote controller and generate appropriate event.	
Feature	• Sample driver for IR remote controller
Limitations	• This is not a general-purpose remote controller driver. You can use the source code of this driver for reference only.

Video Codec

MPEG4 Decoder

MPEG4 Decoder is able to decode various types of MPEG4, such as DivX and XviD. This CODEC supports only decoding. System layer is not included.	
Feature	● DirectShow transform filter ● Supported formats : DivX 3.11, DivX 4.x, DivX 5.x (SP and ASP), XviD Max. resolution : D1 (720x480) Max. bitrate : 3.5Mbps Max. frames per second : 30fps ● Output format : 2D YUV
Performance	● See MPEG4 Decoder performance chart
Limitations	● Do not support 4MV + QPEL contents. ● Do not support GMC which has 2 or more warping points. ● In some DivX 3.11 contents you may see defections

WMV Decoder	
WMV Decoder is able to decode WMV7, 8 and 9 contents This CODEC supports only decoding. System layer is not included.	
Feature	● DirectShow transform filter ● Supported formats : WMV7, WMV8, WMV9 Max. resolution : QVGA (320x240) Max. bitrate : 500kbps Max. frames per second : 30fps ● Output format : 2D YUV
Performance	● QVGA, 500Kbps, 30frames per second
Limitations	Do not support B frame

Audio Codec

Microsoft MP3 Decoder	
MP3 Decoder is able to decode MP3 audio stream.	
Feature	● MP3 Audio Decoder

MES MP3 Decoder

MES MP3 Decoder evaluation version is included in BSP.

Feature	• MES MP3 Audio Decoder Eval.
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WMA Decoder

WMA Decoder is able to decode Windows Media Audio stream.

Feature	• Windows Media Audio Decoder
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Audio Capture

Audio capture filter receives audio input data for encoding.

Feature	• Audio Capture Source Filter • Capture from the microphone input • Fixed sample rate : 22050Hz
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Media File Format

AVI

Feature	• Microsoft AVI File Format
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MPG

Feature	• MPEG1 File Format
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WMV

Feature	• WMV File Format
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ASF

Feature	• ASF File Format
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Other Media Components

DST AVI DEMUX	
Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Demux Microsoft AVI File into audio and video stream

DST AVI MUX	
Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Audio and Video mux filter

DST AVI Writer	
Feature	- Developed by DST Corporation. (Tel. +82-31-710-8800) - Writes AVI multiplexed stream into a file

Sample Applications

MESPlayer	
MES Player is a simple media player for MPEG4 decoding and encoding functions.	
Feature	- Play MPEG4 media file by using MPEG4 video decoder filter - Encode camera input to MPEG4 media file.

OSImageUpdate	
OSImageUpdate is a simple application program that replaces OS image in NAND flash with new image.	
Feature	- Replace OS image with nk.nb0 file in ATA Drive. - Supports OS image file of ROM image format (nk.nb0) - Source code is in APPS\OSImageUpdate folder.

Development Environment

TCP / IP Debugging

Remote debugging tools provided in development environment such as eMbedded Visual C++ or Windows CE Platform Builder can be connected to MMSP2 system through TCP/IP networking.

Feature	• Supports remote debugging of applications that runs on MMSP2 system through TCP/IP network (with eMbedded C++) • Remote debugging tools are available. (ex. Remote process viewer)
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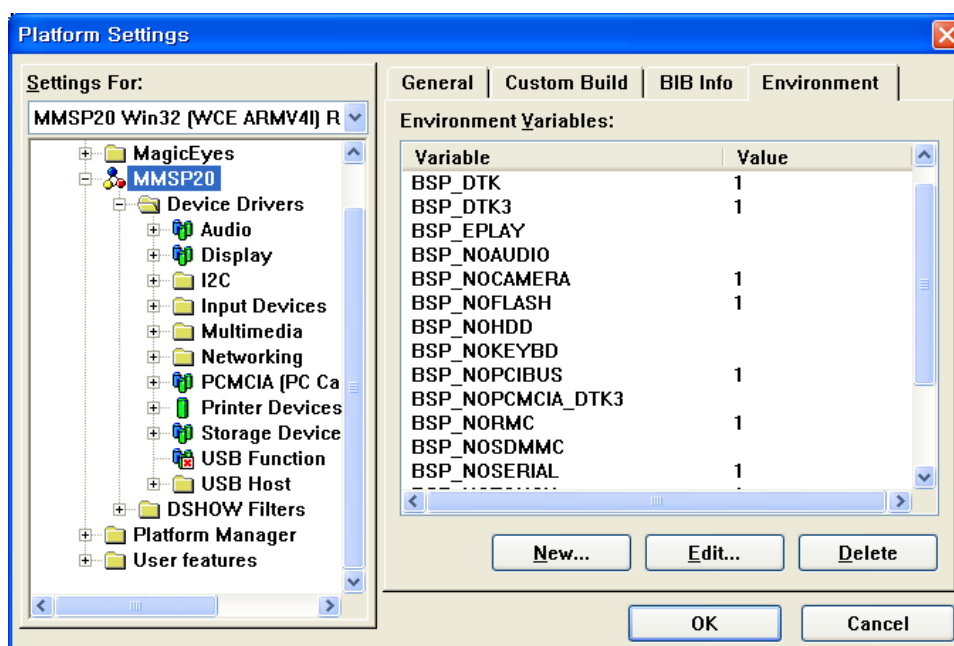
KITL Debugging

Kernel level remote debugging through TCP/IP networking.

Feature	• Supports remote debugging of kernel that runs on MMSP2 system through TCP/IP network • Remote debugging tools are available. (ex. Remote process viewer)
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2. Installation Attention

- To avoid conflict, delete previous BSP before install current BSP.
- Import new .cec file into Platform Builder's **Manage Catalog Features** after delete previous .cec.
- Refer "**BSP Build Guide For MP2520F**" chapter of **MP2520F Development Kit User's Manual** document for information about making new platform and adding necessary components.
- Select **Platform** -> **Settings** from Platform Builder menu and add these variables to Environment tab.



Environment variable setting for each board type is shown below:

H/W	Variables						BSP_SDRA
		BSP_DTK	BSP_DTK3	BSP_DTK4	BSP_PMP	BSP_EPLAY	M_128
Old DTK (before rev.3)		1					1 (null for 64MB)
Old DTK rev.3		1	1				1 (null for 64MB)
PMP2nd					1		
DTK rev.4.1.1, Modified PMP 1.5.3				1			1 (null for 64MB)

- For default LCD output, set DIP switch on the board as:
 - “Off Off Off Off” for DTK 4th board
 - “On On On On” for PMP2nd board
 - “On On Off Off Off Off Off Off” for DTK 3rd or earlier DTK board

3. New Components

3.1. Quick Rebooting Mode (from 2.4.2 alpha)

- Supports quick rebooting mode
- Ext0 button lets the device go into suspend mode. And Ext1 button wakes up the device with quick booting mode turned on. At quick rebooting mode MMSP2 skips reading flash memory. It is booted from Windows CE image remaining in SDRAM.

3.1.1. How to use quick reboot mode

- It only works at DTK4 board which has Ext0 and Ext1 buttons.
- Add environmental variables BSP_QREBOOT as 1, BSP_NOPWRCTRL as null.
- New eBoot have to be written into NAND flash.
- After booting use Ext0 button for suspend mode and Ext1 button for quick reboot.

3.2. Power Suspend Mode (from 2.4.2 alpha)

- At power suspend mode every operation is held. If you push Ext0 button DTK4 goes into power suspend mode.
- Then Ext1button makes system resume previous job.

3.2.1. How to use quick rebooting mode

- It only works at DTK4 board which has Ext0 and Ext1 buttons.
- Environmental variable BSP_NOPWRCTRL should be null.
- Ext0 button is assigned for suspend mode and Ex1 for wake-up

3.3. HS-MMC (from 2.4.2 beta)

- HS-MMC in 4-bit mode is supported.
- Operation clock is 25MHz.
- We have not done compatibility test.
- Only Samsung HS-MMC cards are tested.

3.4. Button device driver

- Source code at MMSP20/DRIVERS/REMOTECTLR.
- Sample application at MMSP20/APPS/ ButtonTest

- Supports **PMP2** and **DTK4** development kit that has 5 general purpose keys (SW3 ~ SW7)

3.4.1. Known problems

- MENU event is not sent to application.

3.5. JPEG Decoder

- JPEG decoder is provided as source filter type
- You can find sample usage in MESPlayer source code

3.5.1. Known problems

- Compatibility is not good.
- We have not tested enough

Modified Components

3.6. AVI DEMUX (from 2.4.2 alpha)

Commercial version of AVI DEMUX is available. This filter is developed by DST Corporation

Version 1.5

- removed audio noise which occurs in the case that the length of audio chunk is zero

Version 1.40

- fixed the stop problem at the start position when we play some files like CD1_320x240_Divx5_24fps_780k_MP3_22K_40k.avi

(www.dstcorp.com)

3.7. MES Player ver. 0.56 for DTK4

MES Player is a simple media player for MPEG4 decoding and encoding functions.

An execution file is included in BSP. You can find the latest source codes on MagicEyes Developer's forum.

Source codes and document are also included in BSP. They are located at MMSP20\APPS\MESPlayer folder.

See ReleaseNote.txt in MESPlayer folder.

3.8. MES MP3 Decoder (from 2.4.2 beta)

- Evaluation version is included without time-limitation.
- Version information is displayed before starting decode.
- Memory leakage problem has been fixed.

3.9. Display Driver

- A bug that cannot read DIP S/W value has been fixed. (from 2.4.2 alpha)
- Supports HITACHI WVGA(800x480) LCD, DIP S/W 0 (from 2.4.2 alpha)
- Clears video memory at booting time (from 2.4.2 alpha)
- Scaler or FDC is turned off after turning off MLC YUV layer (from 2.4.2 beta)

3.10. ISP filter (from 2.4.2 beta)

- A bug that set stream time with wrong value has been fixed

3.11. Audio Driver (from 2.4.2 alpha)

- A bug that don't playback 16/8-bit mono & 8-bit stereo has been fixed.

3.12. Audio Capture Source Filter

- The number of buffers and size of a buffer is optimized. (from 2.4.2 alpha)
- A bug that this filter works only at 22KHz sampling rate has been fixed. (from 2.4.2 beta)

3.13. eBoot (from 2.4.2 alpha)

- supports KITL
- UART baudrate has been changed to 115200bps

3.14. ATA Driver

- Supports HDD on static bus only (from 2.4.2 alpha)
- Sets static bus timing parameters according to the PIO mode of HDD (from 2.4.2 beta)

3.15. Net2272 USB 2.0 Driver

- Not only HDD but also CF, NAND can be used as mass storage. (from 2.4.2 alpha)
- Performance is slightly increased about 5%. (from 2.4.2 beta)

3.16. Memory Configuration for 64MB system (from 2.4.2 alpha)

- reserved more memory space for frame buffer (9MB)
- extended MPEG memory for WMV full performance. (12MB)
- Free memory that application use is reduced to 9MB. To give more space to application reduce Windows image size. Currently memory reserved for Windows image is 30MB.

3.17. Memory Configuration for 128MB system (from 2.4.2 beta)

- extended MPEG memory for WMV full performance. (12MB)
- Free memory for application is increased
- There is no unreserved memory now. We assigned all memory except for HW into free memory.

3.18. ARM940T Device Driver (from 2.4.2 alpha)

- A bug is fixed that release Fast IO semaphore even ARM940T driver don't get it.

3.19. UART (from 2.4.2 beta)

- UART Tx had a bug that lost some data. It is fixed.

3.20. 'Video In' Application (from 2.4.2 beta)

- Supports changing the location of overlay surface in build option

3.21. Playwnd2 Application (from 2.4.2 beta)

- don't generates runtime error when AC3 filter is not included in Windows CE image.

3.22. MESMP4D (from 2.4.2 beta)

- Memory leakage problem has been fixed
- Supports 'xvid' also. (The four characters describing video codec type is placed at the front part of AVI file)

4. New Environment Variables

4.1. BSP_NOPWRCTRL (from 2.4.2 alpha)

- if 1 Power Suspend mode and Quick Reboot mode are disabled.
- If null Power Suspend mode or Quick Reboot mode is enabled.
- See table 1

4.2. BSP_QREBOOT (from 2.4.2 alpha)

- If 1 Quick Reboot mode is enabled
- See table 1

BSP_NOPWRCTRL	BSP_QREBOOT	Mode
null	Null	No power suspend mode, No quick reboot
null	1	No power suspend mode, No quick reboot
1	Null	Power suspend mode
1	1	Quick reboot mode

Table 1 Power suspend and quick reboot mode selection

5. Known Problems

5.1. USB 2.0 Mass Storage

- To use HDD as mass storage you must remove other storages from device before boot-up.
- It means that storage having device name as 'DSK1' is assigned as mass storage. It is decided at boot time.

5.2. MESPlayer

- If you open contents several times MESPlayer may be unstable.

5.3. USB external storages

- There are compatibility issues. Windows CE may not recognize some thumb drives.
- We have not tested USB external HDD.

6. CODEC Status

6.1. MPEG4 Decoder

6.1.1. Supported Video Stream Formats

DivX 3.11, DivX 4.x, DivX 5.x SP, DivX 5.x ASP, XviD

- Don't support Video clips with 4mv and QPEL.
- Don't support GMC with 2 or more warping points.

6.1.2. Performance

DivX 4.x, DivX 5.x SP, XviD : 720x480, 5Mbps, 30fps

DivX 5.x ASP : 720x480, 3.5Mbps, 30fps

DivX 3.11 : 640x352, 1.8Mbps, 24fps

(for maximum performance you should use MES MP3 decoder and MES Player)

6.1.3. Known Problems

- In some DivX 3.11 contents you may see defections
- Video clips without audio may have some problems, irregular playback speed, unable to repeat, etc.
- In some video clips has very long audio chunks, you may see irregular playback after seeking.
- We have been fixed memory leakage problems in MPEG4 decoder at BSP 2.4.2 beta but it has still problem. We asked Microsoft about it. Because even we use all components provided by Microsoft it still shows memory leakage.

6.2. WMV Decoder

6.2.1. Supported WMV Versions

WMV7, WMV8, WMV9

6.2.2. Performance

QVGA (320x240), 500kbps, 30fps

6.2.3. Known Problems

none

7. Release History

7.1. MMSP 2 BSP Release version 1.0 (April 2004)

Following features are supported:

- Windows CE 4.2 Kernel
- Display driver
- USB Host
- USB Function
- AC97
- ATA HDD
- NAND flash storage
- CS8900 Ethernet
- PCMCIA (source only)
- MPEG4 Hardware accelerated codec

7.2. MMSP 2 BSP Release version 2.0 (May 2004)

- Codec is improved.
- Stability of ATA HDD driver is improved.
- Performance issues related to display drivers are improved.

7.3. MMSP 2 BSP Release version 2.2.0 (June 2004)

- SD/MMC memory card driver is included.
- Camera driver that supports 1.3 mega pixel camera module is included.
- HDD recognition failure problem is addressed.
- Frequency of noise occurrence is reduced.
- Stability of long-time play is improved.

7.3.1. MMSP2 BSP Release Version 2.2.1 (July 2004)

- System time update failure problem is addressed.
- Decoding failure problem with some XviD media file is addressed.
- Audio codec initialization failure problem that occurs at times is addressed.
- USB keyboard (HID) is recognized correctly.

7.3.2. MMSP2 BSP Release Version 2.2.2 (September 2004)

- AC3 Codec is supported
- Failure of USB function driver is addressed.
- Performance of DivX 3.11 is improved.
- DMA problem of AC97 is addressed
- Register writing problem is fixed.

7.3.3. MMSP2 BSP Release Version 2.4.0 (November 2004)

- Audio capture source filter is added
- MPEG4 Encoder is added
- MPEG4 supports B frame decoding
- Evaluation version of the AVI MUX and the AVI SPLITTER is included (developed by DST)
- System timer failure occurred by the bug of the on chip timer is addressed. (After bug is fixed, only one of four timers on MMSP2 chip is available. This timer is used for scheduling.)
- System crash problem that occurs during audio volume control is fixed
- Compatibility of the ATA driver is improved
- New catalog settings for video-in components

7.3.4. MMSP2 BSP Release Version 2.4.1 (May 2005)

- MESPlayer : a bug not release overlay surface has been fixed
- MESPlayer source codes and Document are included.
- Video capture shows correct pictures.
- Removed useless delay in Net2272 device driver
- Includes new ARM940T firmware that has MP4D, MP4E, H.263D, H.263E, WMV, MP3D. DivX 3.11 performance has been improved.
- Documents for DST components-MUX, DEMUX,WMV- are included.
- 64MB Memory configuration has been changed to increase 1D memory.
- ARM940T device driver source codes is included.
- DST player lite is included.
- Reduced MBParam buffer size from 2MB to 1MB.
- DST DEMUX ver. 1.2
- WMV ver. 1.1
- Moves NET2272 driver at final stage of booting sequence to avoid 'unknow device error' on PC.
- eBoot loading time has been reduced.
- MES MP3D evaluation version is included. It has time-limit about 2min 30seconds.
- A bug of MESMP4D filter that not release ARM940T device driver handle has been fixed.